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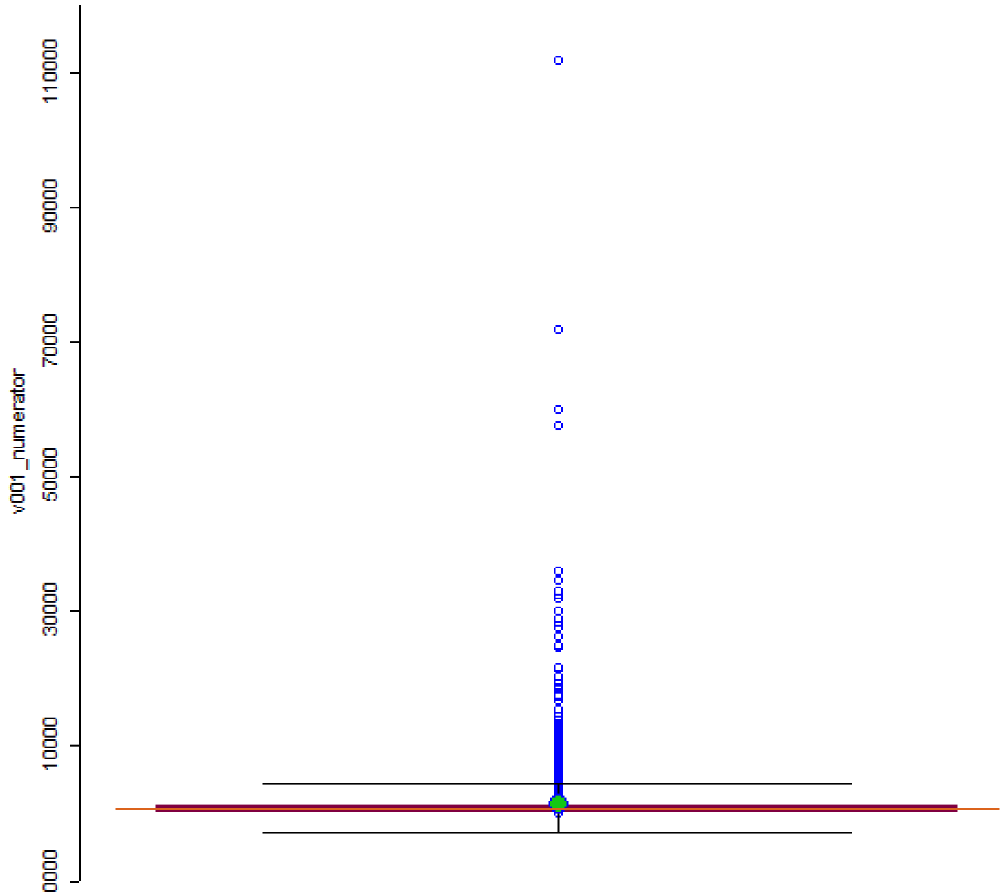
HW8 – GeoDa
Advanced Vector GIS
03/13/2026

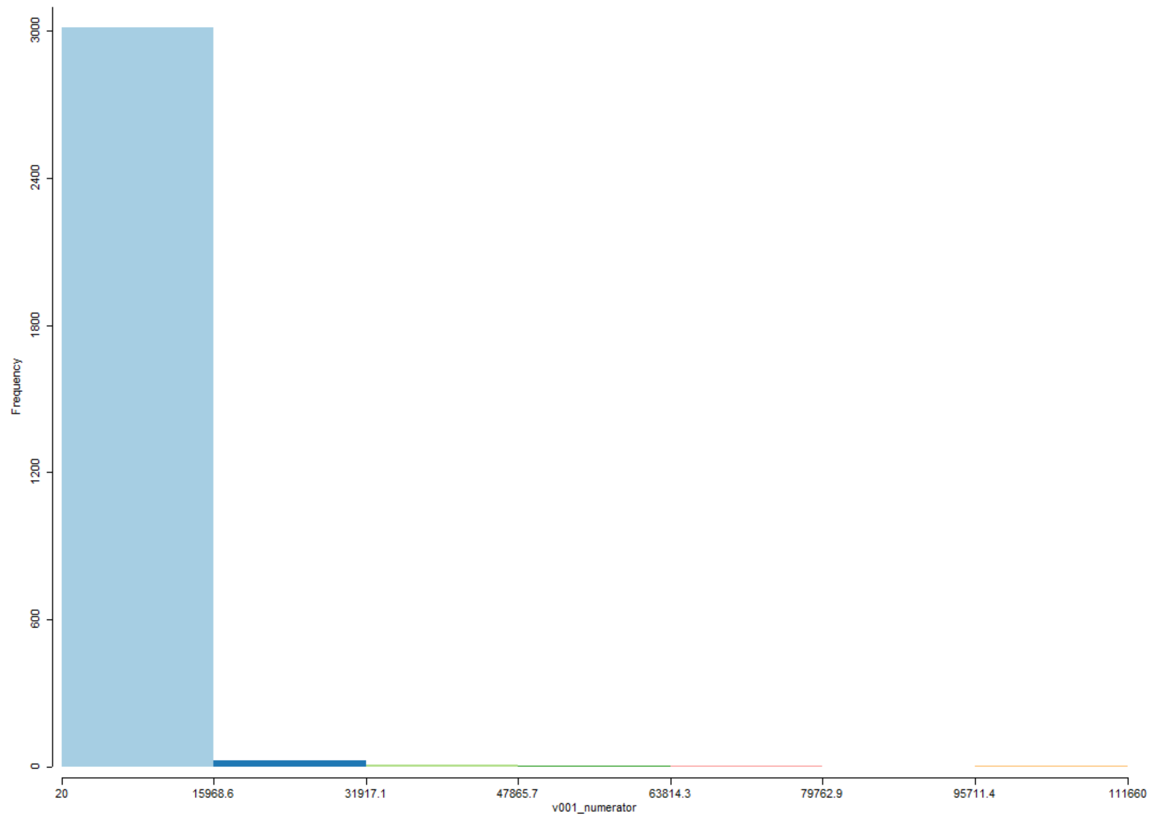
Two variables from County dataset:

1. Percentage of adults 18-64 without insurance
    a. v003_rawvalue
2. The numerator is the cumulative number of years of potential life lost from deaths among county residents under age 75, over a three-year period.
    a. v001_numerator

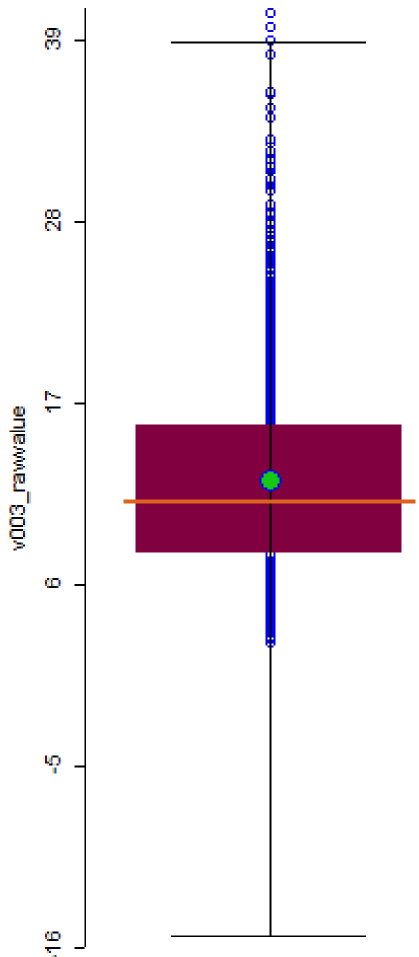
Q1) I chose to use the v003_rawvalue variable (Percentage of adults 18-64 without insurance) and the v001_numerator variable (cumulative number of years of potential life lost from deaths among county residents under age 75). I think these two variables may have a significant relationship because in the USA, having health insurance is usually a strong indicator for a person's willingness/ability to seek medical care that might prevent early (or preventable) death. I think the number of people that don't have insurance will be a predictor for number of years of potential life lost. There are 243 outliers in the v001 variable (with hinge = 3) and 3 outliers in the v003 variable (with hinge = 3).



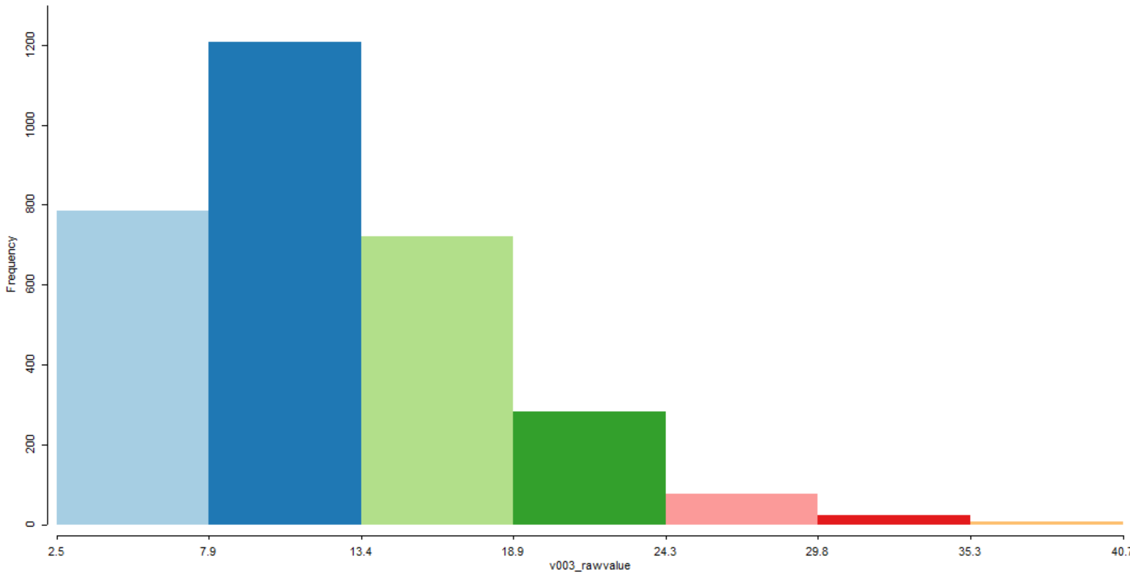
	v001_numerator
min	20
max	111660
Q1	230
median	520
Q3	1255
IQR	1025
mean	1538.5490
s.d.	4148.3744



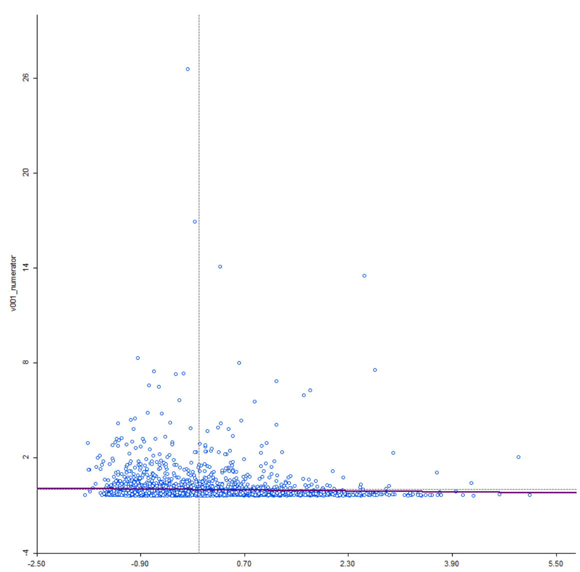
3000
2400
1800
1200
600
0
Frequency
20
15968.6
31917.1
47865.7
63814.3
79762.9
95711.4
111660
v001_numerator



	v003_rawvalue
min	2.4813
max	40.7318
Q1	7.9017
median	11.0068
Q3	15.6518
IQR	7.7502
mean	12.2801
s.d.	5.6015



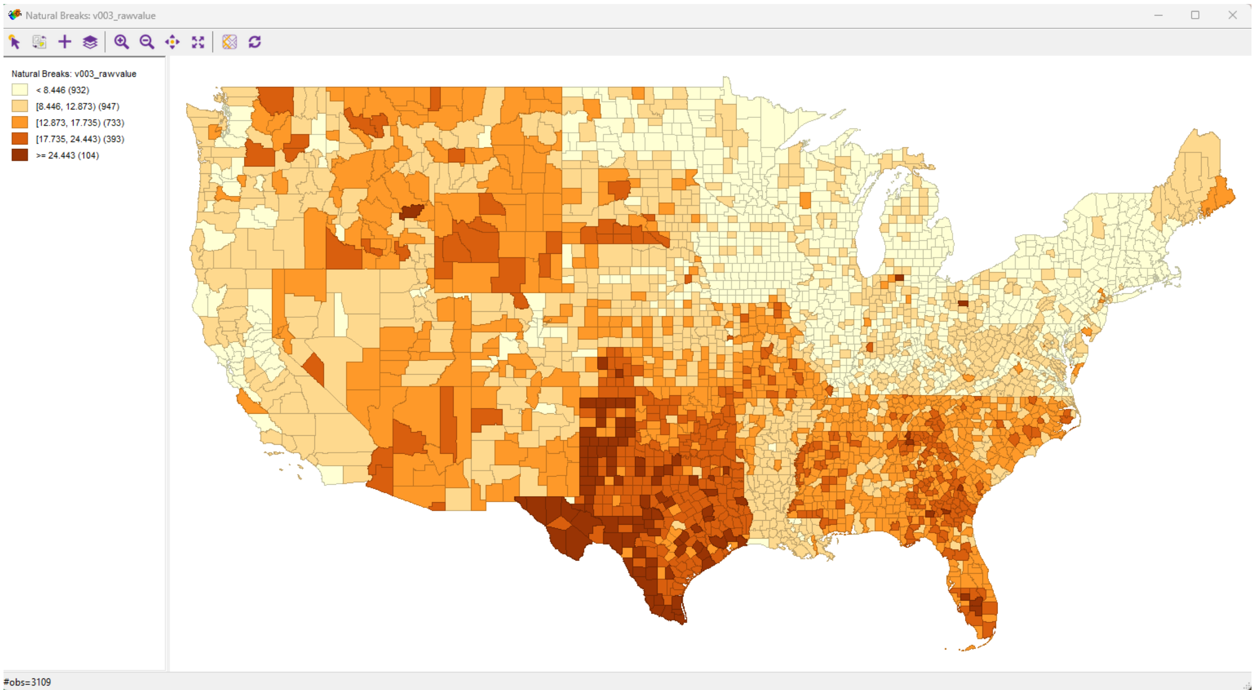
Q2) The correlation between these two variables is negative (-0.037), meaning that as the value of people who don't have insurance goes up, the number of cummulative years of potential life lost goes down.



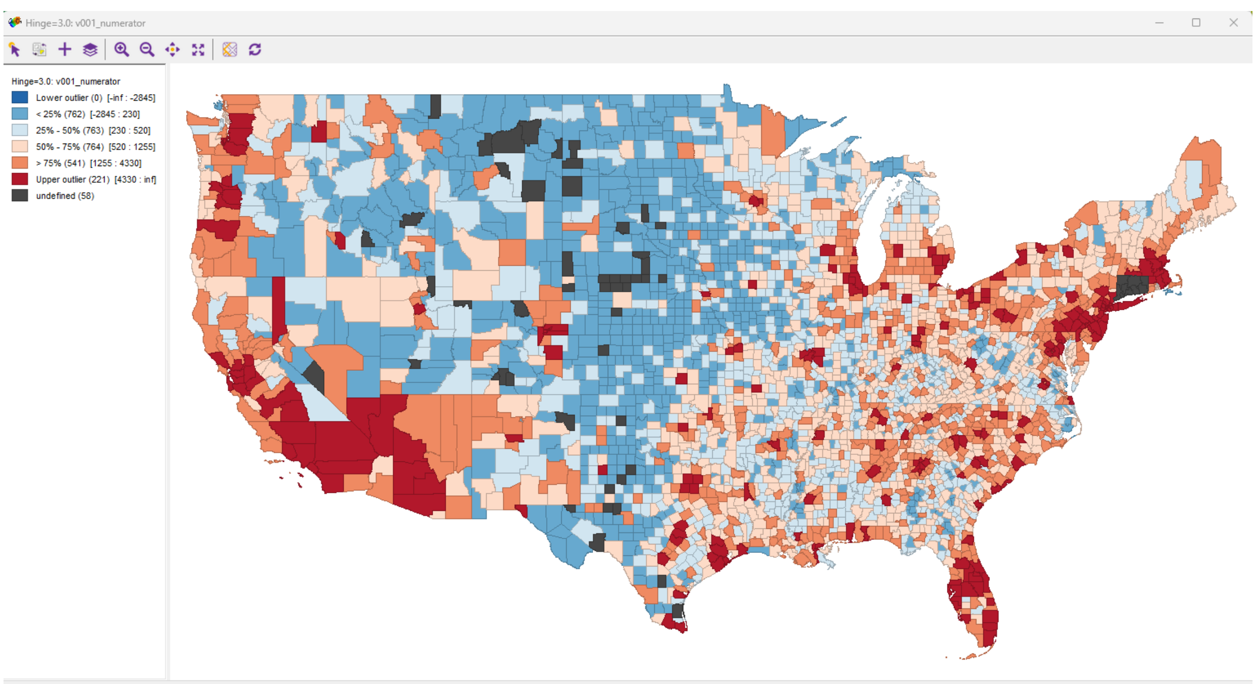
#obs	R^2	const a	std-err a	t-stat a	p-value a	slope b	std-err b	t-stat b	p-value b
3051	0.001	0	0.018	0	1	-0.037	0.018	-2.038	0.042
0	0	0	0	0	0	0	0	0	0
3051	0.001	0	0.018	0	1	-0.037	0.018	-2.038	0.042

Chow test for sel/unsel regression subsets: need two valid regressions

Q3) For the v003_rawvalue variable, I chose the natural breaks map because I think this variable has significant right skew, but not a significant amount of outliers. Unlike the other variable, there were a very small number of upper outliers, so I think this map does a good job of visually and spatially displaying the relative distribution.

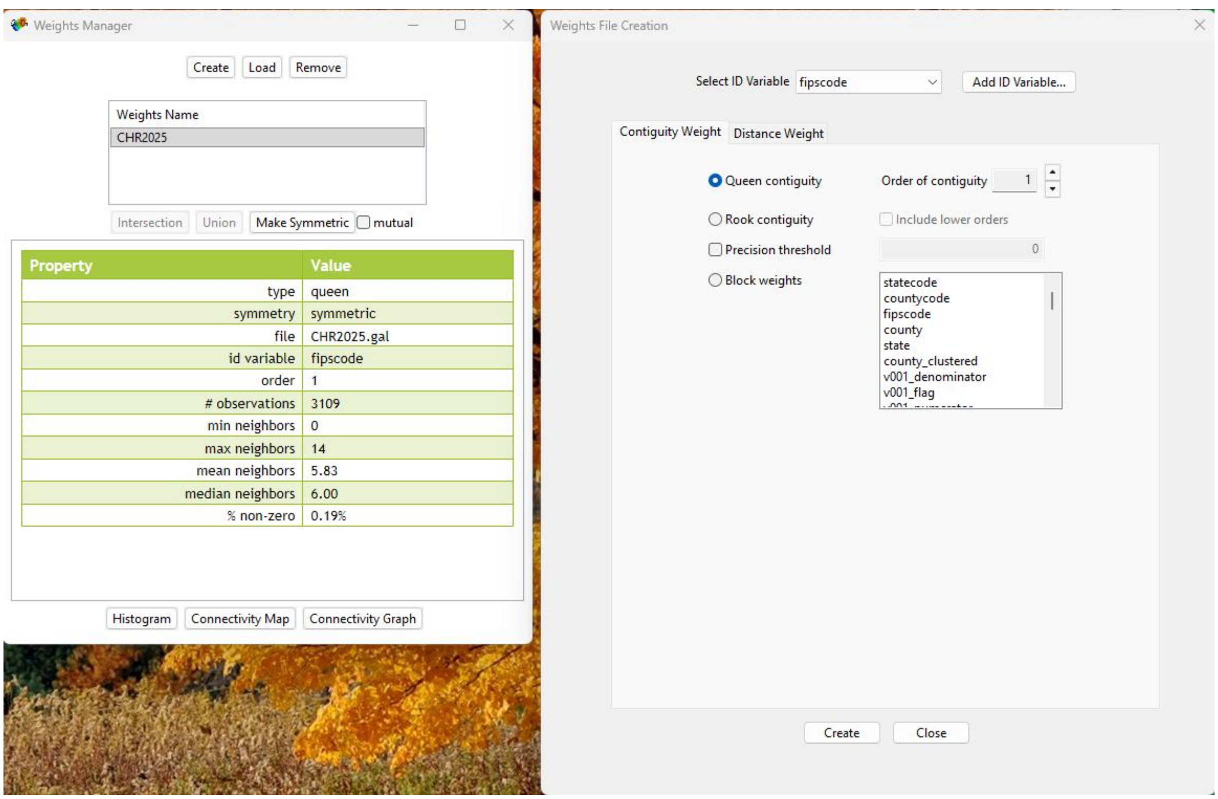


I decided to use the box map with hinge = 3 for the v001 variable because the distribution has a significant amount of upper outliers and I wanted to see them clearly as well as their spatial distribution.

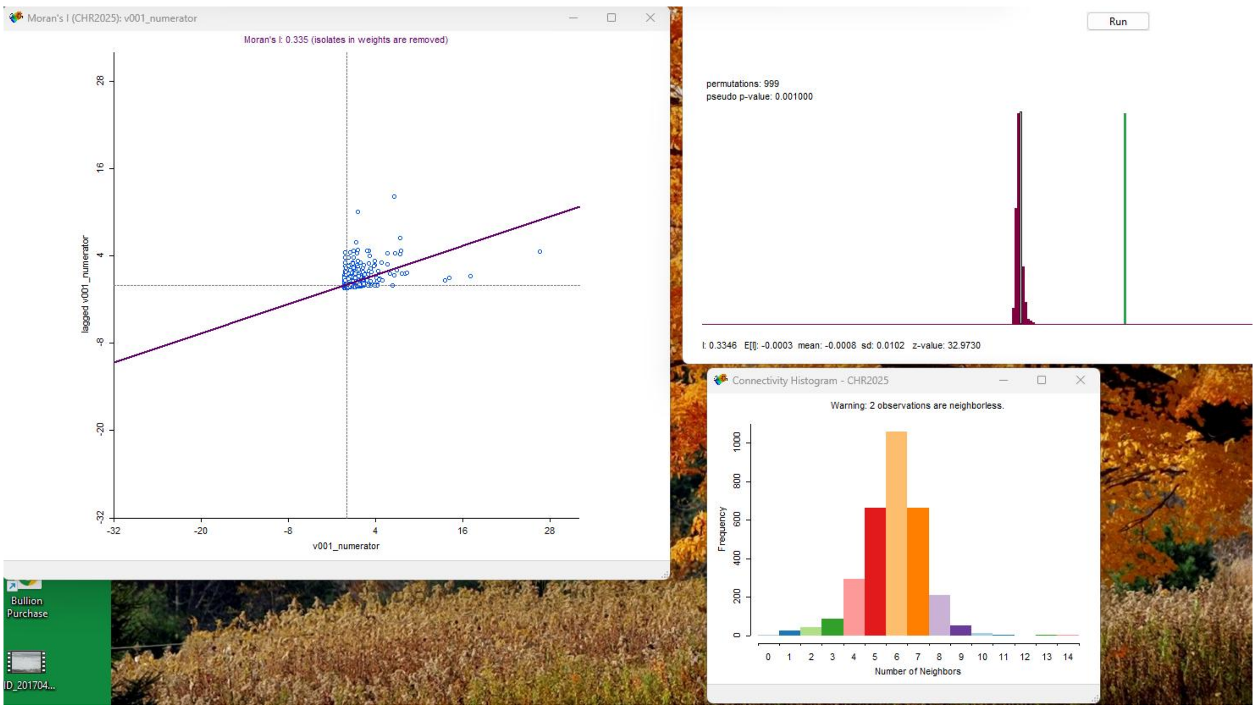


Q)4 – Q)8: Results of the Univariate Moran and Univariate Local Moran's I and repsonse:

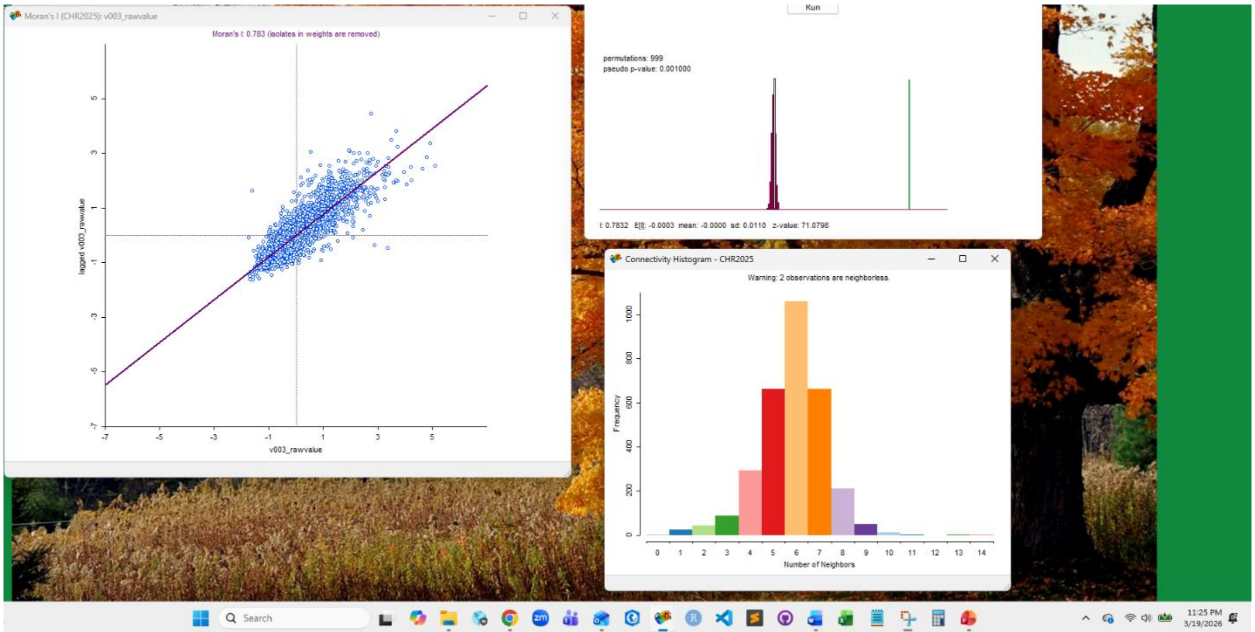
Before running the Univariate Moran's I and Univariate Local Moran's I, I created a weight file. Because this dataset is county polygon file, and the variables I am mapping are number of uninsured people and number of potential years of life lost, I decided that contiguity weight makes the most sense. Considering the fact that a discrete administrative boundary such as county deliniations may also imply certain policy differences that can impact a resident's ability to have insurance, or conditions that may lead to residents loss of potential life, I think distance makes less sense. Queen continguity will provide a better sense of potential auotcorrelation because I want the analysis to include all adjacent counties. Lastly, I think first order of contiguity is good because there are many counties, and I don't want to diminish the tools ability to detect local patterns in the data.



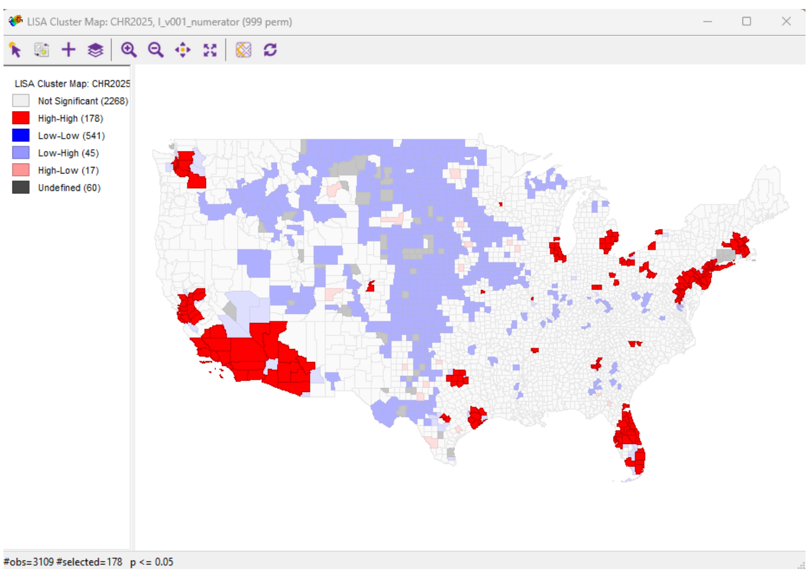
For the v001 variable, the value of the univariate Moran's I is 0.335. This value is positive but not large (not close to 1), so I know there is a slight positive spatial correlation, meaning similar values are closer to each other than not. I also know that this value is statistically significant because when I randomize this model with 999 permutations the pseudo p-value is 0.001 and z-value is 32.97. This means there is high certainty that this is not a random result, and I can reject the null hypothesis.

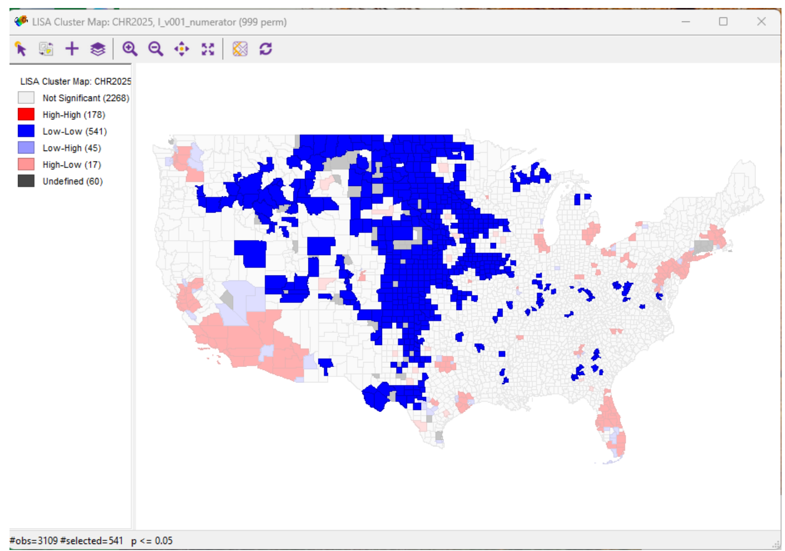


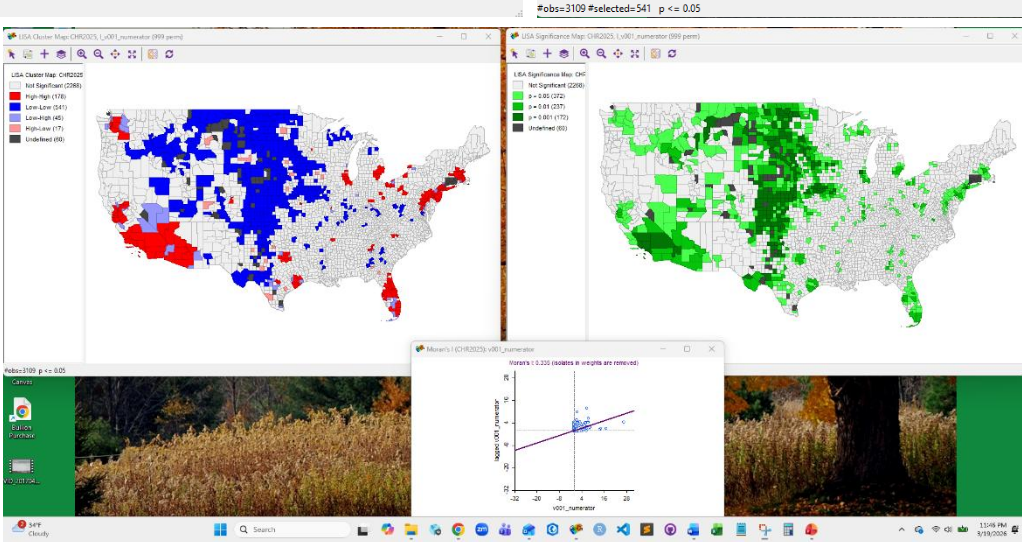
For the v003 variable, the value of the univariate Moran's I is 0.783. This value is positive and much closer to 1 than the other variable, meaning there is a positive correlation with more significant clustering over similar values. I know that this value is statistically significant because when I randomize the model with 999 permutations the pseudo p-value is 0.001 and the z-value is 74.8951. This means there is high certainty that this is not a random result, and that I can reject the null hypothesis.



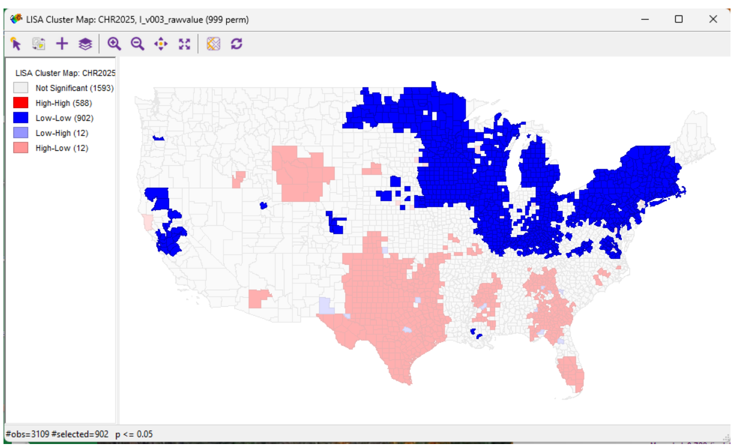
For the v001 variable, the Local Moran's I (LISA) clusters are mainly Low-Low (other than non-significant). The Low-Low cluster is mostly in the center of the country, with less intense clustering to the east and west of center. This dark blue area means the features have lower values than other features, and their neighborhoods have lower values than other neighborhoods. The second highest cluster type is High-High which has two main concentrations - one in the northeast coast and the other in the southwest coast. There are less intense clusters of High-High in the eastern half of the country. These High-High clusters mean that these features have higher values than other values, and their neighborhoods have higher values than other neighborhoods. The Low-High clusters are the third most numerous category with significantly less features than the first two. These features are located mostly on the west coast but have some scattering across the eastern half of the country as well. The Low-High clusters mean the features have lower values than other features, but the neighborhoods have higher values than other neighborhoods. Lastly, the least numerous category is High-Low, which are mainly spread throughout the center of the country, throughout the main cluster of Low-Low features. This would make sense, because this is the area with the highest concentration of low value neighborhoods in the country. High-Low clusters mean the feature value is higher than other features, but the neighborhood values are lower than other neighborhoods.

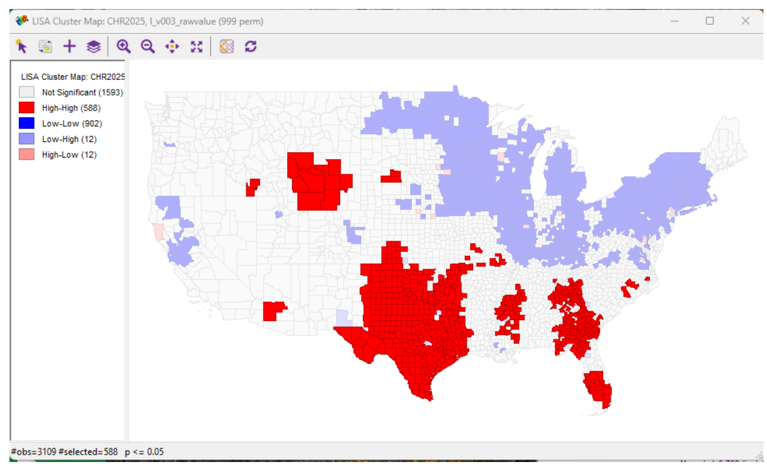


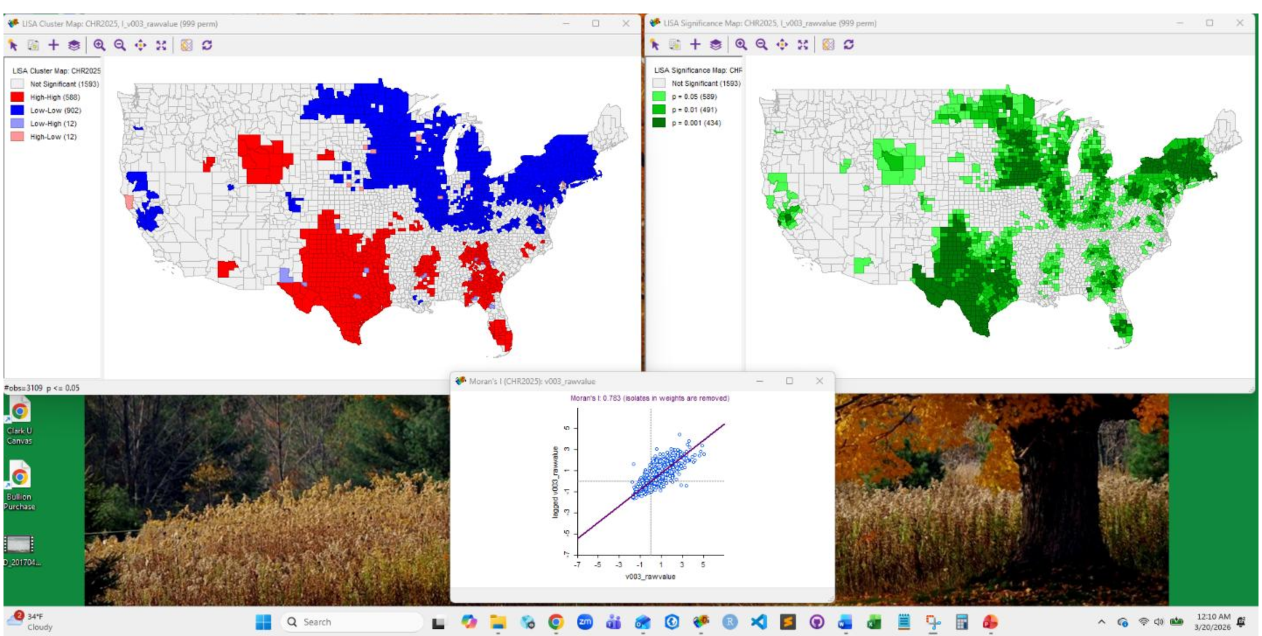




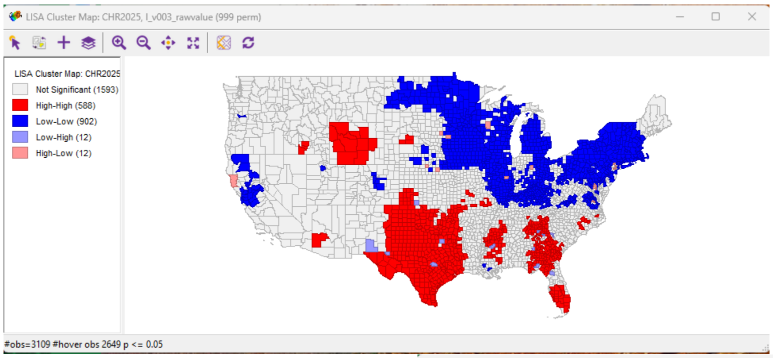
For the v003 variable, the Local Moran's I (LISA) clusters are mainly Low-Low (other than non-significant). The main Low-Low cluster spreads across the the northeast into the north midwest with high intensity. This means these features have lower values than other features, and their neighborhoods have lower values than other neighborhoods. The second most numerous cluster type is High-High. The most intense clustering of these values is happening in Texas and Oklahoma, with other grouped clustering areas in Wyoming, and in the south east. These features have higher values than other features and their neighborhoods have high values than other neighborhoods. Both Low-High and High-Low clusters have the same number of features. The Low-High features are scattered around the High-High cluster areas, and these features have lower values than other features, but their neighborhoods have higher values than other neighborhoods. Conversley, the High-Low features are scattered through the Low-Low cluster area. These features have higher values than other features and their neighborhoods have lower values than other neighborhoods.

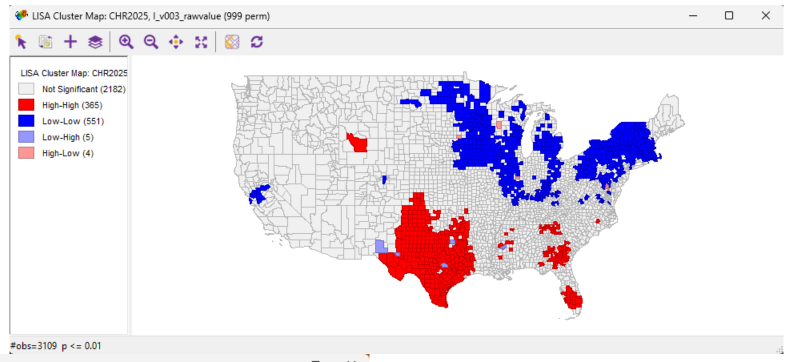


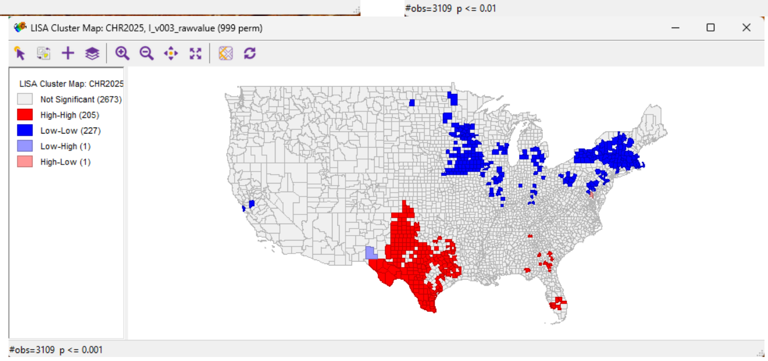




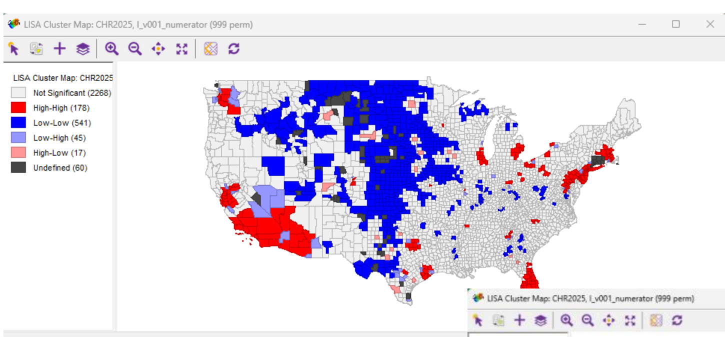
For the v003 variable, changing the significance level from .05 to .01 to .001 reveals that most of Texas remains significant with High-High clusters. Likewise, the northeast and the sizable area in the Midwest around southern Minnesota, western Wisconsin, and northern Iowa remain significant Low-Low spots.

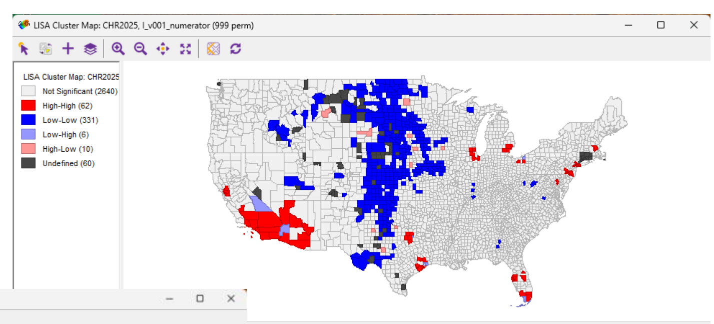


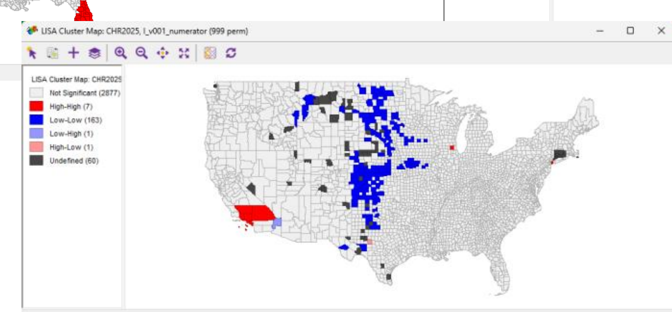




For the v001 variable, changing the significance level from .05 to .01 to .001 reveals that the Low-Low cluster in the center of the country remains significant at all levels. To a lesser extent, there is a small High-High cluster that remains at all levels in southern California.







In conclusion, there was less correlation between these two variables than I thought there would be. In addition, I thought for sure the correlation would be positive, not negative (-0.037) – that is, I thought the relationship would be the more people are uninsured the more loss of potential life in that county would be. But the -0.037 correlation between these variables suggests that the relationship is actually the opposite: the more people are uninsured the less loss of potential life in that county. That being said, the correlation is so close to zero that the relationship is almost negligible. However, there is a clear statistical significance of spatial clustering happening in both variables. In addition, I can reject the null hypothesis for both variables (that there is spatial randomness). Overall, the most numerous and prevailing clustering trend is Low-Low - most clustered features have a low feature value and their neighborhoods are also low.